MILITARY SPECIFICATION

TIMER, SEQUENTIAL: M42E2

This amendment forms a part of Military Specification MIL-T-45886A (AR) dated 25 May 1988, and is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1 Specifications and standards, delete: "MIL-I-6866"

PAGE 10

3.13 Soldering, delete sentence and replace with "All soldering shall be in accordance with MIL-STD-2000 Task G."

PAGE 11

Add new paragraphs 3.16, 3.17, and 3.18 as follows:

"3.16 Safety critical component control (SCCC) program.

3.16.1 Critical component. In accordance with Nuclear Weapon Systems Safety Group (NWSSG) report 87-6 (SECRET), the LANCE M42E2 Timer has been determined to be a Nuclear Safety Critical Component of the LANCE Missile System.

3.16.2 Control of components. The SCCC program requires that the nuclear safety critical components within the M42E2 timer or those components associated with the acceptance of the M42E2 timer be controlled to preclude inadvertent or deliberate modification of the software computer program codes located within the M42E2 timer and the Acceptance Inspection Equipment (AIE). The components associated with the SCCC program that must be controlled are as follows:

Masked ROM within M42E2 Timer	P/N 12551085
Acceptance Inspection Equipment (AIE)	P/N 8809650
AIE Validated Master Program code disks	

It is mandatory that the computer program code and all memory storage locations be certified such that said codes and memory locations are as specified in drawing package documentation. The contractor must develop and implement a manufacturing process flow procedure which will at some point in the manufacturing process flow validate the computer code/complete

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memory storage locations (both used and unused) in each production M42E2 timer and control the access to the M42E2 timers and AIE from that point until the arrival of the M42E2 timers at first military destination as specified in the contract.

3.17 Safety critical component control requirements.

3.17.1 Verification. The contractor must verify at some point in the manufacturing process that all memory locations in the M42E2 Masked ROM P/N 12551085 are exactly as specified in drawing 12550186. From the point of last complete verification the contractor must implement personnel handling and manufacturing safeguards to insure the validated integrety of the Masked ROM IC(s) located within each M42E2 timer until arrival of timers at first military destination.

3.17.2 Test equipment control. The contractor will take receipt of government furnished AIE equipment. Equipment will be delivered and set-up by the government and validated as to the integrety of the embedded computer program and all unassigned memory locations exactly as required by drawings 8809651 and 8809652. Tamper Detection Seal(s) (with controlled serial numbers) will be applied by government personnel after the validation process. The contractor must then develop and implement personnel control procedures to insure that the AIE equipment (firmware and software) has not been compromised.

3.17.3 Master program code. The contractor will be provided under serparate courier service, a certified master program code for both the M42E2 timer masked ROM and the AIE via program listing and AIE disk respectively. Both items are the masters against which the production timer assemblies and the AIE must be validated. As such the contractor is to store these items in an approved container capable of securing classified information at the SECRET level. Removal of the master program code from the container will require two person control at all times. Use of the program codes to perform ROM cross-checks or formatting (re-formatting) the AIE will require two personnel, each of whom can detect unauthorized actions (inconsistent, inadvertent, or deliberate actions with respect to approved procedures). At no time will the master program codes be available to a single individual.

3.17.4 Production process. Likewise the production process involving the masked ROM P/N 12551085 must be under two person control once the ROM has been validated against the master controlled program code associated with the ROM. Both persons must be capable of detecting unauthorized actions. At all points throughout the production process (piece pelt, installed on printed circuit board, installed in open frame timer assembly, undergoing environmental testing, etc.) the ROM IC must be secure.

3.18 Tamper seals.

3.18.1 Procedure. The contractor will be provided Tamper Detection Seals (serialized and controlled) to use in securing containers, storage cabinets, environmental chambers, etc.) Procedures for application, removal and

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identification of any tampering will also be provided. Tamper Detection Seals must be secured as classified material and must be handled (dispensed, applied, removed, etc) via two knowledgeable personnel (as defined previously). Contractor shall be instructed on how to apply seals, seal removal, cleaning and inspection procedures. The contractor in turn must provide for all personnel involved with this effort a training program and ongoing awareness program on security requirements and inspection criteria that apply.

3.18.2 Personnel clearance. All personnel involved with M42E2 timer production or AIE equipment must have a minimum clearance of confidential.

3.18.3 Associated personnel and cost. The contractor must determine and provide the additional personnel and/or additional hours associated with this control process."

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4.4.2.5 Add the following test items as follows:

"Major:

109.	Verify ROM obtained from security container (SC) visual.	Table IV	3.2
110.	Verify ROM control number from operation sheets visual.	Table IV	3.2
111.	Verify storage in SC visual. "	Table IV	3.2

PAGE 22

Add new paragraph 4.4.2.7.1 (included as page 22a).

PAGE 25

4.4.2.9 Switch, optical, Major 106: delete "3.16" and substitute "3.15".

PAGE 30

4.4.2.15:

Major category 102: delete "four" and substitute "eight".

Major category 110: delete in its entirety.

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4.4.2.16, Major category:

112: Delete in its entirety.

Add the following:

"116.	Verify electronic assemblies were obtained from security container	Table IV	3.2	Visual
117.	Verify application of tamper seal	Table IV	3.2	Visual
118.	Verify record of tamper seal control number	Table IV	3.2	Visual"

PAGE 32

Add new paragraph 4.4.2.18 (included page 32a).

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.7.1	Electronic Assembly	1	1	12551084
				NEXT HIGHER ASSEMBLY 12551040

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical:	None defined			
Major:				
101.	Verify printed board assemblies obtained from security container (SC)	Table IV	3.2	Visual
102.	Verify storage in SC	Table IV	3.2	Visual
Minor:	None defined			

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	ROM Microprocessor	1 OF 1	12550185
			NEXT HIGHER ASSEMBLY 12550186

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical:	None defined			
Major:				
101.	Verify program validation	Table IV	3.2	Visual
102.	Verify record of control number	Table IV	3.2	Visual
103.	Verify storage in security container (SC)	Table IV	3.2	Visual
Minor:	None defined			

NOTES:

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Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1135-A334)

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